

Discussion 13.1 4-bit Ring Counter

Course Code: CPE 013

Program:

Course Title: Logic Circuits and Design

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Section: 22S2

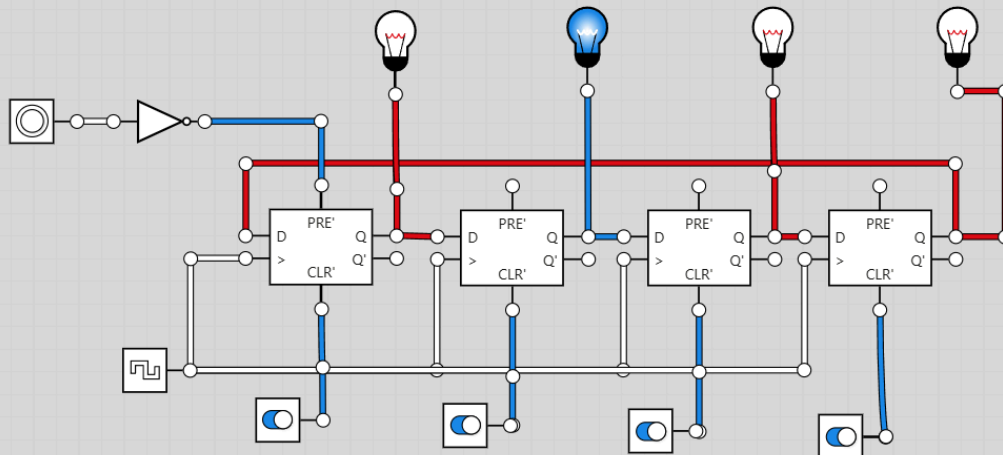
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Instructions

Answer



In my simulation of the 4-bit Ring Counter, I observed that the circuit requires an initial "preset" pulse to load a single high bit into the first flip-flop (FFA). Because the PRE and CLR pins are active-low, I had to keep these switches at Logic 1 for the counter to function properly; otherwise, the outputs remained frozen. Once the clock was activated, I saw the single "ON" state shift sequentially from QA to QB, then QC, and finally QD with each clock pulse. The feedback loop from the last flip-flop back to the first ensured the bit continuously recycled, creating a repeating 4-state sequence where only one output was active at a time.